

Huiwen Alex Yang

University of California, Berkeley

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Education

University of California, Berkeley

Berkeley, CA, USA

PhD in Psychology, with a Designated Emphasis in Cognitive Science

2023 - 2028 (expected)

- Advisor: Dr. Celeste Kidd
- Committee: Dr. Celeste Kidd, Dr. Bill D. Thompson, Dr. Steve T. Piantadosi, Dr. Alison Gopnik

University of California, Berkeley

Berkeley, CA, USA

BA in Computer Science, Physics, Psychology

2017 - 2021

- Distinction in general scholarship

Publications

Published

Yang, H.A., Ho, M.K., Thompson, B.D. (2026). Hypothesis-guided discovery of cognitive algorithms via program refinement. *arXiv preprint*.

Yang, H.A., Kidd, C. (2026). What certainty and other metacognitive representations in cognitive development reveal about the future of AI models. *DevAI Workshop at NeurIPS 2026*.

Yang, H.A., Skwersky, R., Thompson, B.D., Kidd, C. (2026). AYO: A Safe and Privacy-Preserving Platform for Studying Children's Interactions with LLMs. *Child Safety in AI Workshop at NeurIPS 2026*.

Yang, H.A., Ho, M.K., Thompson, B.D. (2026). Spontaneous meta-learning of efficient problem-solving algorithms. In *Proceedings of the 48th Annual Meeting of the Cognitive Science Society*.

Yang, H.A., Thompson, B.D., Kidd, C. (2025). Children spontaneously discover efficient solutions to a difficult sorting task. In *Nature Human Behaviour*.

Fang, Y., **Yang, H.A.**, Christie, S. (2025). Early experiences in shaping children's explore-exploit decisions: evidence from the rural-urban gap. In *Proceedings of the 47th Annual Meeting of the Cognitive Science Society*.

Yang, H.A., Martí, L., Baer, C., Granera, A., Palmeri, H., Kidd, C. (2024). Emergence of certainty representations for guiding concept learning. In *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*.

Yang, H.A., Thompson, B.D., Kidd, C. (2024). Children spontaneously discover efficient sorting algorithms in a seriation task. In *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*.

Yang, H.A., Piantadosi, S.T., Kidd, C. (2023). Children's estimation of peripheral information drives improvements in approximate number sense. In *Proceedings of the 45th Annual Meeting of the Cognitive Science Society*.

Awards, Fellowships, & Grants

2026	Graduate Division Conference Travel Grant, UC Berkeley	\$ 1,500
2025	Berkeley Economy & Society Initiative Technology Collaborative Grant, UC Berkeley — with Sarah Stolp Shepherd, Evan Orticio & Celeste Kidd	\$ 2,500
2024	Graduate Division Conference Travel Grant, UC Berkeley	\$ 1,500
2024	Departmental Award, UC Berkeley Graduate Diversity Program	\$ 500
2023	Departmental Award, UC Berkeley Graduate Diversity Program	\$ 500
2023	Berkeley Fellowship for Graduate Study, UC Berkeley	\$ 34,000

Presentations

Conference Presentations

Yang, H.A., Skwersky, R., Thompson, B.D., Kidd, C. (2026). AYO: A Safe and Privacy-Preserving Platform for Studying Children's Interactions with LLMs. Oral presentation: Child Safety in AI Workshop at NeurIPS 2026, Atlanta, USA.

Yang, H.A., Martí, L., Baer, C., Granera, A., Palmeri, H., Kidd, C. (2024). Emergence of certainty representations for guiding language learning. Oral presentation: the 46th Annual Meeting of the Cognitive Science Society, Rotterdam, the Netherlands.

Yang, H.A., Martí, L., Baer, C., Granera, A., Palmeri, H., Kidd, C. (2023). Emergence of certainty representations for guiding language learning. Oral presentation: ProtoLang Conference, Rome, Italy.

Invited Talks

Yang, H.A. (2026). Certainty Representations and Algorithmic Discovery as Mechanisms of Intrinsic Motivation. ARIA (NSF AI Research Institute on Interaction for AI Assistants).

Yang, H.A. (2025). Behavioral experiments and computational models: essential tools for advancing cognitive science. Guest Lecture: Psych 147: Methods in Cognitive Development, University of California, Berkeley.

Yang, H.A. (2024). Computational models in cognitive developmental research. Guest Lecture: Psych 147: Methods in Cognitive Development, University of California, Berkeley.

Posters

Conference Posters

Yang, H.A., Kidd, C. (2026). What certainty and other metacognitive representations in cognitive development reveal about the future of AI models. Poster: DevAI Workshop at NeurIPS 2026, Atlanta, USA.

Yang, H.A., Ho, M.K., Thompson, B.D. (2026). Spontaneous meta-learning of efficient problem-solving algorithms. Poster: the 48th Annual Meeting of the Cognitive Science Society, Rio de Janeiro, Brazil.

Yang, H.A.*, Liu, Q.*, Thompson, B.D. (2025). Human-level cross-sensory mappings from language alone. Poster: the 66th Annual Meeting of the Psychonomic Society, Denver, USA.

Fang, Y., **Yang, H.A.**, Christie, S. (2025). Early experiences in shaping children's explore-exploit decisions: evidence from the rural-urban gap. Poster: the 47th Annual Meeting of the Cognitive Science Society, San Francisco, USA.

Sheperd, S., **Yang, H.A.**, Kidd, C. (2025). Bright shiny garbage: Video content shown to low-income children is characterized by higher flicker. Poster: the 47th Annual Meeting of the Cognitive Science Society, San Francisco, USA.

Yang, H.A., Thompson, B.D., Kidd, C. (2024, 2025). Children Spontaneously Discover Efficient Sorting Algorithms in a Seriation Task. Poster: the 46th Annual Meeting of the Cognitive Science Society, Rotterdam, the Netherlands. Poster: 15th annual Budapest CEU Conference on Cognitive Development (BCCCD), Budapest, Hungary.

Yang, H.A.*, Liu, Q.* , Thompson, B.D. (2024). Why Banana is like a Cloudy Day: Cross-domain mappings in Human and Large Language Models. Poster: the 5th International Conference on Analogy, Amsterdam, the Netherlands.

Yang, H.A., Piantadosi, S.T., Kidd, C. (2023). Children’s Estimation of Peripheral Information Drives Improvements in Approximate Number Sense. Poster: the 45th Annual Meeting of the Cognitive Science Society, Sydney, Australia.

Teaching Experience _____

- Fall 2026 Psych 1: General Psychology, Teaching Assistant
- Spring 2024 Psych 147: Methods of Cognitive Development, Teaching Assistant

Additional Training _____

- 2025 Diverse Intelligences Summer Institute (DISI), University of St Andrews
- 2023 Workshop on Computational Cognitive Models of Learning and Development, Harvard University

Research Experience _____

University of California, Berkeley - Dept of Psychology - Kidd Lab *Berkeley, CA, USA*
Advisor: Dr. Celeste Kidd and Dr. Steven Piantadosi *Jul 2021 – Aug 2023*
• Research focus: Approximate number system, Attention and gaze, Algorithm acquisition, Certainty, Cognitive loads, Perceptions/bias

Tsinghua University - Dept of Statistics - Deng’s Lab *Beijing, China*
Advisor: Dr. Ke Deng *Jun 2019 – Aug 2019*
• Research focus: Statistic machine learning, Image classification, Neural network, Generalized zero-shot learning, Imbalanced dataset

ICSI (International Computer Science Institute) *Berkeley, CA, USA*
Advisors: Dr. Stella Yu *Nov 2019– Mar 2020*
• Research focus: Image reconstruction and classification, Dictionary learning, Sparse coding, Point set dataset

Service _____

Journal Reviewing
Journal of Memory and Language (2025)

Conference Reviewing

Cognitive Science Society (2023 - 2026)

Professional Affiliations

Cognitive Science Society (2023 -); Psychonomic Society (2025)

Work Experience

Overeality Labs

Mountainview, CA, USA

Founding Team Member, VP of Product & Assistant Tech Lead

Jun 2021 – May 2022

- Co-founded an NFT incubator that promotes DAO-selected NFTs to bring crypto culture into the Metaverse
- Led the product team that consists of 4 product designers and 9 programmers that designed and implemented the Web3 platform, Marketplace and various Web2 components
- Worked with solidity engineers and constructed a layer 1 blockchain
- Coordinated with the marketing team and the cryptography research team on designing features that best suit the needs of our product
- Acquired 130000 users in 6 months and raised 5 million dollars during the seed round

Alibaba Group

Beijing, China

Frontend Software Engineer Intern

Dec 2018 – Jan 2019

- Developed a web-based data visualization platform using React and Node.js
- Helped the frontend developing team track their project progress, control their product pipelines and communicate with other software engineering teams

Mentorship

Research Collaborators

Joseph Ong (2026–Present), URAP Student

Thresia Vazhaeparambil (2024–2026), Honors Thesis Student

Aaraam Granera (2023–2024), UX Researcher, Coinbase

Research Assistants

Shawn Sikder (2026–Present), Undergraduate; **Hanieh Shirvani** (2026–Present), Undergraduate; **Caleb Gilbert** (2026–Present), Undergraduate; **Lauren Felder** (2025–Present), Undergraduate; **Stacy Kim** (2025–Present), Undergraduate; **Lily Nguyen** (2025–Present), Undergraduate; **Julia Chen** (2025–2026), Undergraduate; **Darshita Prathap** (2025), Undergraduate; **Karina Murugesu** (2024–2025), Undergraduate; **Sarah Benedict** (2024–2025), Undergraduate; **Sean Bullock** (2023–2024), Graduate Researcher, San Jose State University; **Gabrielle Cooke** (2023–2024), PhD Student, Palo Alto University; **Lucy Ko** (2022–2023), Graduate Student, UCLA.